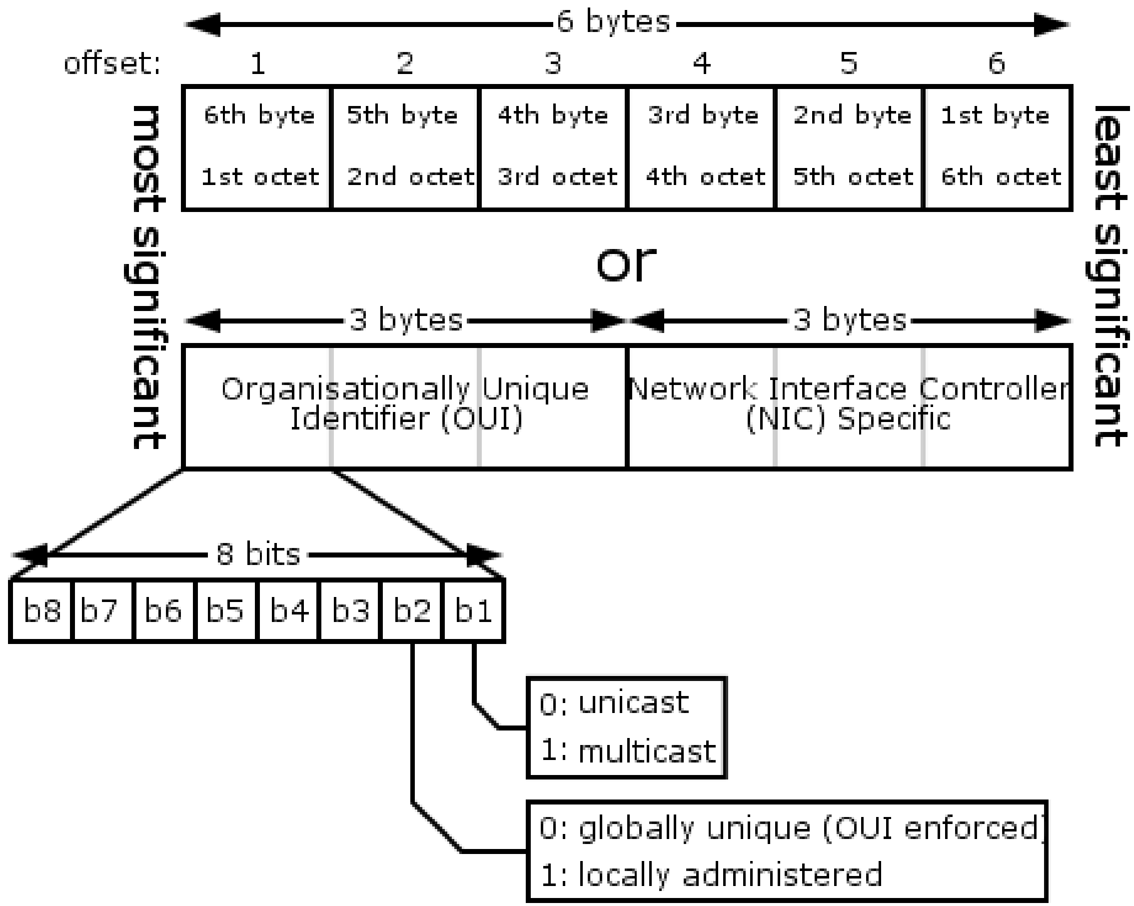
Physical addressing

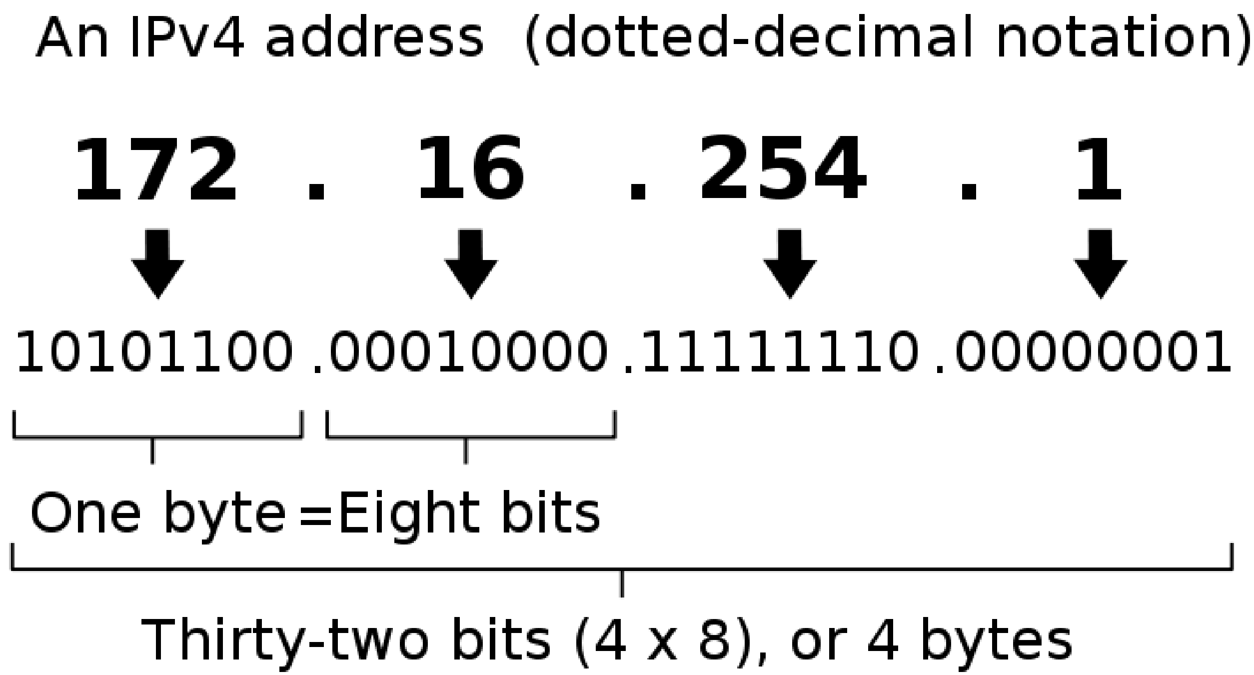


Source: wikipedia.org

IEEE 802.3 Ethernet frame fields

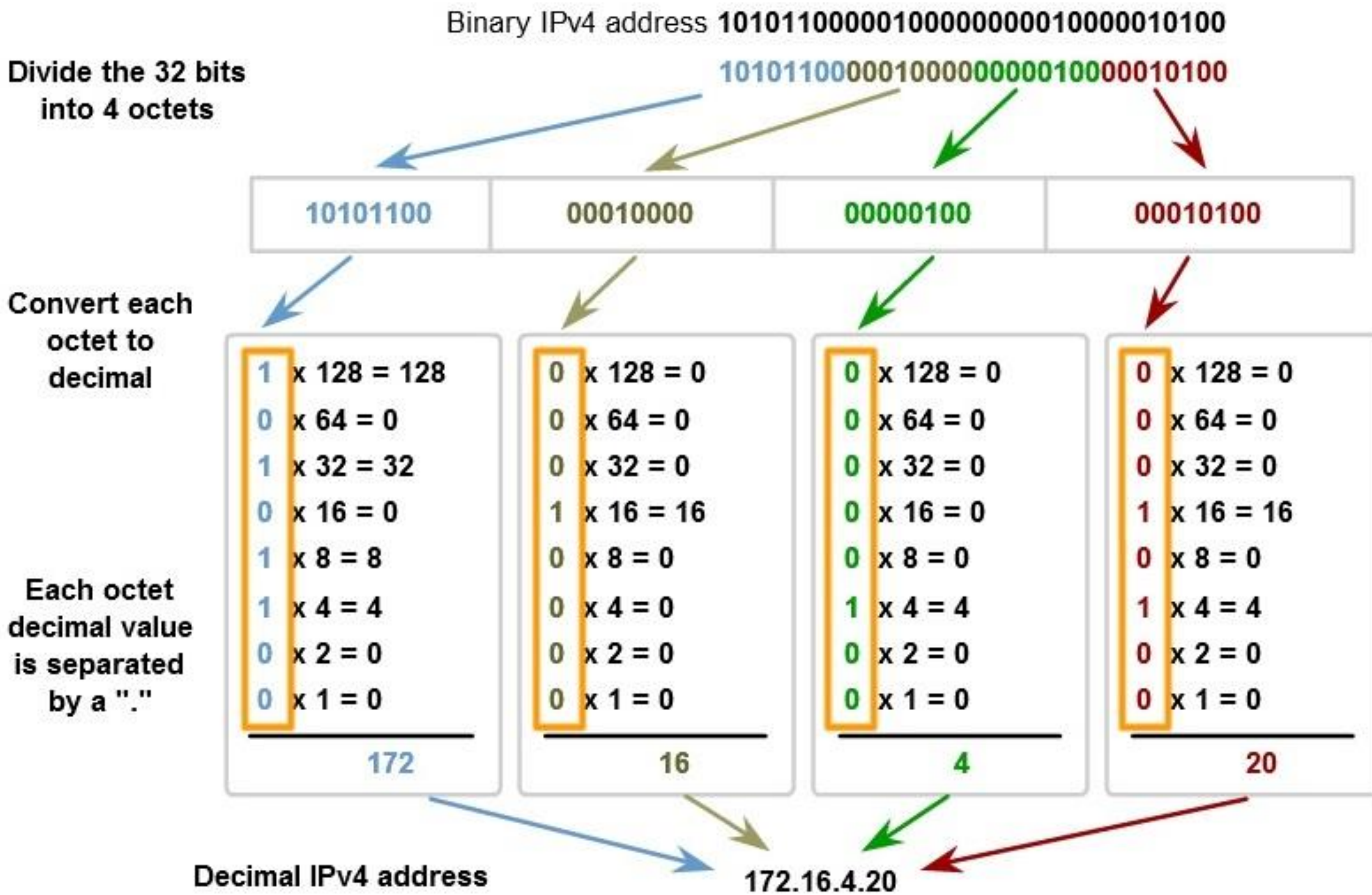
Preample	Frame start	Dest. MAC address	Source MAC address	Lenght/ Type	Encapsulated DATA	FCS
7	1	6	6	2	45-1500 Bytes	4

The structure of IP addresses



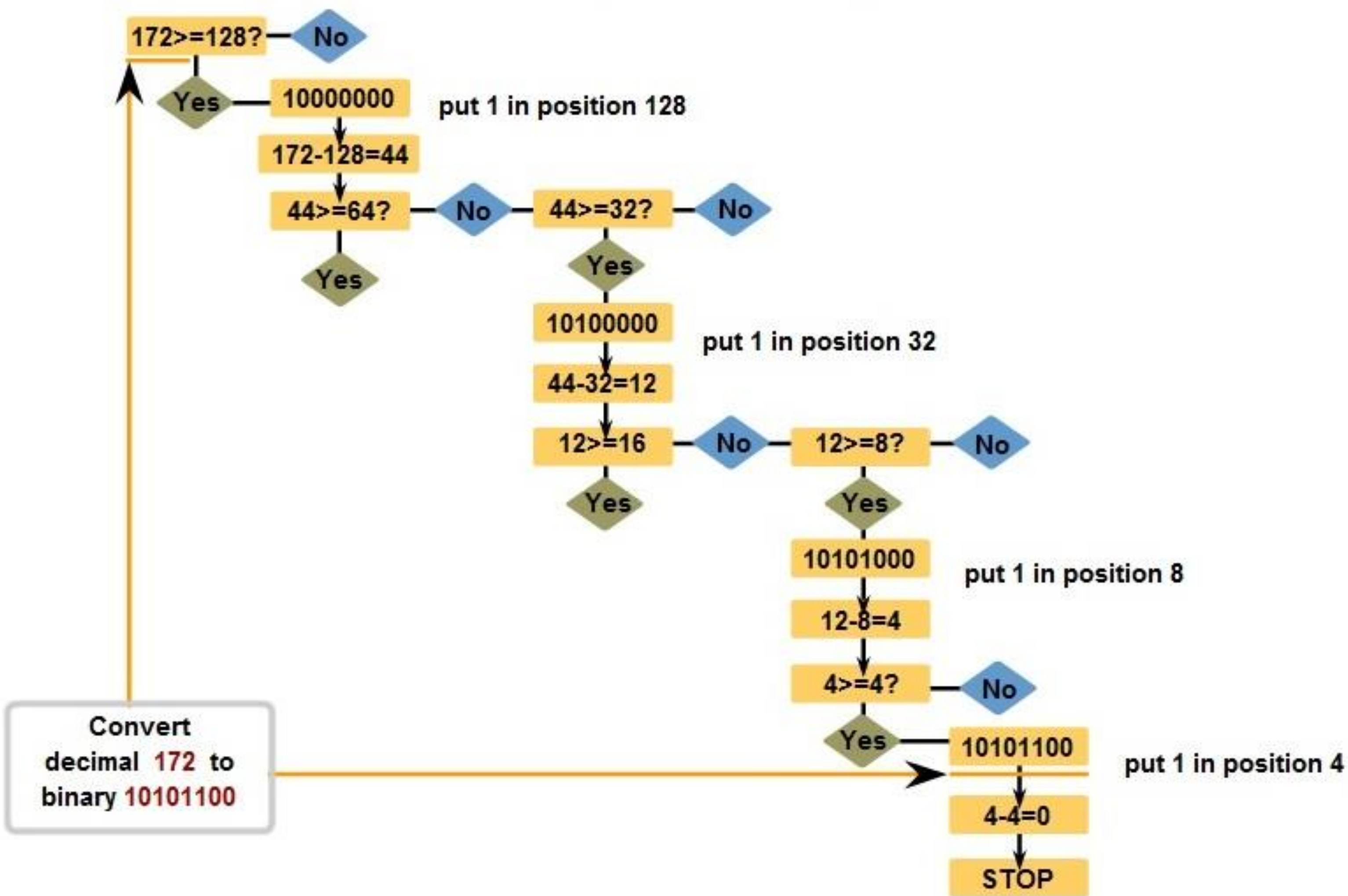
Source: wikipedia.org

Binary to Decimal IP address conversion



Source: highteck.net

Decimal to binary IP address conversion



Source: highteck.net

IPv4 address classes

Address class	I. octet range	I. octet bits	Network and host portions of an address	Default subnet mask
A	0-127	**0**0000000- **0**1111111	**N.**H.H.H	**255.**0.0.0 **11111111.**00000000.00000000.00000000
B	128-191	**10**000000- **10**111111	**N.N.**H.H	**255.255.**0.0 **11111111.11111111.**00000000.00000000
C	192-223	**110**00000- **110**11111	**N.N.N.**H	**255.255.255.**0 **11111111.11111111.11111111.**00000000
D	224-239	**1110**0000- **1110**1111	Multicast címek!	
E	240-255	**1111**0000- **1111**1111	Kísérleti célra!	

Subnet mask

- The mask length is 32 bits and the format is dotted decimal or prefix.
- In the subnet mask, the bit positions for network identification are set to '1' and the positions for host identification are set to '0'.

Address class	Dotted decimal format	Prefix format
A	255.0.0.0	/8
B	255.255.0.0	/16
C	255.255.255.0	/24

Examples:

10.0.0.0 255.0.0.0
152.25.0.0 255.255.0.0
209.46.50.0 255.255.255.0

10.0.0.0/8
152.25.0.0/16
209.46.50.0/24

Special IP addresses

- **Network address:** all node bit positions are '0'
  E.g. 74.0.0.0, 152.10.0.0, 193.6.140.0
- **Directed broadcast address:** all node bit positions are '1'
  E.g. 74.255.255.255, 152.10.255.255, 193.6.140.255
- **Broadcast address on the current network:**
  all bit positions set to ' 1 ' 255.255.255.255
- **Loopback address:** 127.x.y.z
  127.0.0.1

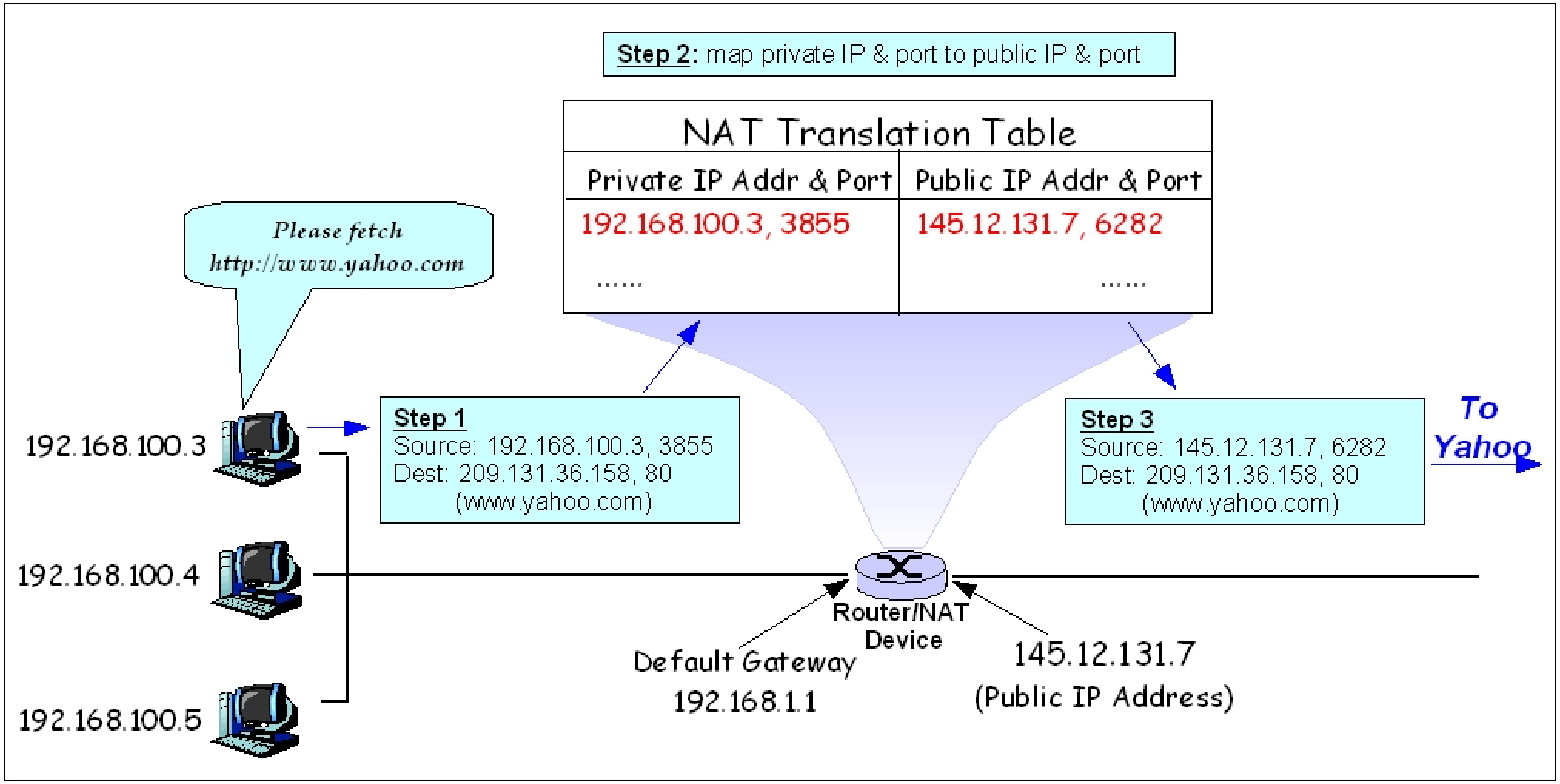
- **Private address ranges** (see next slide)

The network address and the broadcast address cannot be allocated to a node!

Private IP addresses

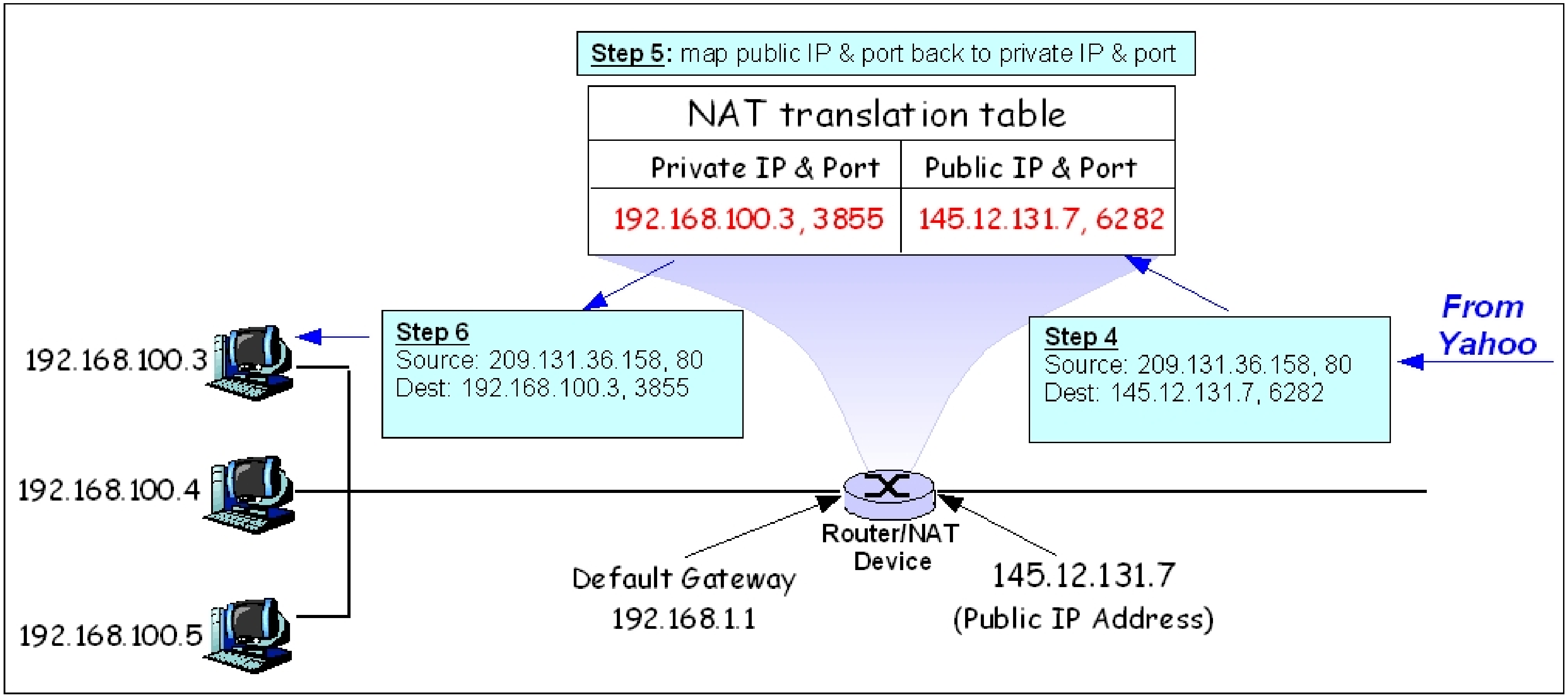
Size	Range	Prefix	Description by class	Largest CIDR block
24-bit block	10.0.0.0-10.255.255.255	/8	1 block of Class A	10.0.0.0/8
20-bit block	172.16.0.0-172.31.255.255	/12	16 continuous Class B blocks	172.16.0.0/12
16-bit block	192.168.0.0-192.168.255.255	/16	256 continuous Class C blocks	192.168.0.0/16

Network Address Translation



Source: en.wikibooks.org

Network Address Translation (continued)



Source: en.wikibooks.org

Tasks

- Calculate the network address associated with the IP address 10.150.0.36 if the subnet mask is 255.0.0.0

- What will be the network's broadcast address?

- Find the first and last assignable address of the network.

Tasks

- Calculate the network address for the IP address 172.16.9.43 if the subnet mask is 255.255.0.0

- What will be the network's broadcast address?

- Find the first and last assignable address of the network.

Tasks

- Calculate the network address associated with the IP address 192.168.10.150 if the subnet mask is 255.255.255.0

- What will be the network's broadcast address?

- Find the first and last assignable address of the network.

Tasks

- Calculate the network address and broadcast address for the IP address 172.19.189.55 if the subnet mask is 255.255.255.224